Learned Controllers for Latent-Preserving Context Switching in Multi-Agent LM Dialogue

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Abstract

We study whether a small, separately-trained controller can teach two frozen language-model agents to switch conversational topic when a third, uninformed agent joins — while still conveying a hidden variable to each other through the switched surface text. Two "insider" agents (A, B) share a latent topic z; a third "outsider" agent C, unaware of z, joins mid-conversation. A lightweight controller module, trained via reinforcement learning while the base language model stays frozen, learns to steer A and B's turns after C's entrance so that B can still recover z while C's recovery accuracy is suppressed. This connects to zero-shot coordination in multi-agent RL and to informal human behaviour (e.g., two friends changing topic when a third joins, without losing their shared meaning). The project trains only a small controller (1–5M parameters) on top of a frozen, quantized 1–3B model runnable on a single laptop GPU/NPU, using a fixed but extensible set of latent topics. We report a measurable accuracy trade-off between B's recovery and C's detection, and a secondary probe into whether the hidden topic remains linearly recoverable from internal representations even when C's black-box guess fails.

1 Problem Statement & Motivation

When a third person joins a conversation about a sensitive subject, speakers often switch topic on the surface while preserving meaning for each other — a common, everyday behaviour with no direct analogue yet studied in language-model agents (e.g., two friends discussing whether to break up with a partner switch to discussing "renovating the apartment" when the partner walks in — "tear it down and start over" still lands as the real message underneath). This resembles the zero-shot coordination (ZSC) problem in cooperative multi-agent RL (Hu et al., 2020; Cui et al., 2021), where two agents must act consistently under a shared latent signal without having been jointly trained for the exact situation they face. Studying this in LM agents connects three under-explored directions: emergent coordination between LM-based agents (Baroni et al., 2022; Riedl, 2025), covert-channel behaviour in natural-language generation (Wu et al., 2024), and adversarial evaluation of what a third party can infer from a transcript. The problem is well suited to a focused RL-style project: it has a clear latent variable, a clear adversary, and a measurable trade-off, without requiring fine-tuning of the base model or a retrieval component.

2 Research Questions

- **RQ1:** Can a small controller, trained on top of a frozen LM, improve an insider agent B's ability to recover a hidden topic z after a context switch, compared to a hand-written topic-switching baseline?
- **RQ2:** Does the learned controller reduce an outsider agent C's ability to guess z from the same transcript, compared to a hand-written topic-switching baseline?
- **RQ3:** Is there a measurable trade-off between B's recovery accuracy and C's recovery accuracy as the adversarial weighting in the reward is varied?
- **RQ4 (secondary):** Is z still linearly recoverable from the controller's internal representations even when C's black-box guess is near chance?

3 Related Work Positioning

This project builds on three research lines: zero-shot coordination in cooperative multi-agent RL (Hu et al., 2020; Cui et al., 2021), where agents must generalize to unseen partners under a shared convention; emergent communication between

learned agents (Baroni et al., 2022; Riedl, 2025), extended here to LM-based dialogue; and work on steganography and covert channels in generated text (Wu et al., 2024), which studies how information can be embedded in surface-plausible output. The reward structure also echoes adversarial neural cryptography (Abadi and Andersen, 2016), where Alice and Bob are trained end-to-end so that a third network, Eve, cannot recover an encrypted message. Our setting differs in three ways: (1) the channel is natural-language dialogue generated by a frozen, pretrained LM rather than raw bit-vectors produced by fully trainable networks; (2) only a small controller and decoder head are trained — the base model's weights and its language capacity are never touched, so any hiding strategy must work through discourse-level topic choice rather than learned low-level encoding; and (3) the adversary C is a fixed, non-co-trained LM prompted to guess a topic from a plausible conversation, not a jointly-optimized adversarial network — so we are measuring robustness to a realistic, non-adaptive eavesdropper rather than to a co-evolving one. It is distinct from prompt-based multi-agent LLM pipelines (negotiation, debate) in that the coordination behaviour is learned by a dedicated module rather than elicited purely through prompting, and distinct from instruction fine-tuning or RAG approaches, neither of which is used here.

4 Proposed Method

Agents. A and B are instances of the same frozen, quantized language model (1–3B parameters, run locally via CPU/GPU-NPU inference). A knows a hidden topic z, drawn from a small set of predefined topic pairs (e.g., a sensitive topic and a plausible cover topic); B does not see z directly but must infer it from A's utterances. C is a separately prompted instance of the same base model with no access to z, used only to guess z from the transcript; C is not trained, avoiding the instability of adversarial co-training.

Controller. A small trainable module (an MLP or single-layer transformer, a few million parameters) observes the dialogue state and z, and produces a conditioning signal for A and B's post-switch turns — either a discrete choice of cover topic or a soft-prompt vector prepended to the frozen model's input (Lester et al., 2021). This is the only trainable component besides a small decoder head on B that predicts z from context. The base language model's weights are never updated.

Training. The controller and B's decoder head are trained with REINFORCE (Williams, 1992) over short multi-turn rollouts, using reward:

$$R = \mathbb{I}[\text{B recovers } z] - \lambda \cdot \mathbb{I}[\text{C recovers } z] \quad (1)$$

This adversarial reward structure follows the Alice–Bob–Eve formulation of Abadi and Andersen (2016), adapted here to a frozen-LM, natural-language setting rather than learned bit-level encryption. B's decoder head is pretrained on non-switched conversations first, so the controller receives a meaningful signal from the start rather than co-evolving with an untrained decoder. A moving-average baseline and an entropy bonus are used to reduce variance and discourage the controller from collapsing to a single cover topic regardless of z.

Scope. The set of latent topics is deliberately kept small and fixed for tractability, but surface phrasing of A's opening turns is varied throughout training (not only at test time) so the controller cannot solve the task by memorizing surface patterns. If time allows, the topic set and cover-topic mappings will be extended beyond the initial fixed set to test whether the learned strategy generalizes, rather than treating this as an in-scope guarantee.

Baseline. As a non-learned point of comparison, we implement a hand-written topic-switching baseline: for each z, a cover topic is pre-assigned via a fixed lookup table, and A switches to it using a small set of scripted transition templates (e.g., "anyway, speaking of [cover topic]…"), with no conditioning on dialogue history and no adaptation of phrasing beyond this static mapping. This baseline is designed to switch topic convincingly enough to already suppress C's recovery of z, so the informative comparison is not whether C is fooled, but whether B's recovery rate under the trained controller exceeds B's recovery rate under this fixed baseline while C's recovery remains equally or more suppressed. This isolates whether learning the switch, rather than merely performing one, drives the accuracy trade-off reported in RQ1–RQ3.

5 Evaluation Plan

- **Primary:** B's z-recovery accuracy, compared against a no-switch ceiling baseline and a no-controller (switch but no training) baseline.

- **Adversarial:** C's z-recovery accuracy under the learned controller versus under a hand-written, non-learned topic-switching strategy, at matched conditions.

- **Trade-off analysis:** B accuracy and C accuracy reported jointly across two to three values of λ, to characterize the trade-off rather than a single operating point.

- **Naturalness (lightweight):** a brief internal check, by the team, on whether switched conversations read as plausible rather than abrupt; not a full human study, given the project's scope.

- **Interpretability probe:** a linear probe trained on the controller's internal representation to test whether z remains recoverable even when C's guess is near chance.

6 Deliverables & Timeline (8 weeks, 4 members)

- **Weeks 1–2:** Define topic set and phrasing templates; build local inference pipeline; run naive-switch baseline against C as an early feasibility check.

- **Week 3:** Implement controller and conditioning mechanism; checkpoint decision on soft-prompt vs. discrete conditioning based on a quick sanity test.

- **Week 4:** Implement and pretrain B's decoder head; implement C's guessing procedure.

- **Weeks 5–6:** Implement and run REINFORCE training across λ values; fall back to a best-of-N selection procedure if RL training does not converge in time.

- **Week 7:** Run interpretability probe (data logging started in Week 5); consolidate baselines and results.

- **Week 8:** Final write-up and internal review.

Final deliverable: A short paper reporting the controller architecture, training procedure, and the B-vs-C accuracy trade-off, together with the interpretability probe results.

7 Expected Contributions

- A lightweight controller architecture for latent-preserving topic switching on top of a frozen language model, trained without modifying base model weights.

- An empirical trade-off analysis between an insider agent's message recovery and an outsider agent's detection, under a controlled adversarial setting.

- A secondary interpretability result on whether hidden information persists in internal representations independent of surface-level detectability.

8 Risks & Limitations

- **Small, fixed topic set:** Limits generalization claims; mitigated by extending the topic set opportunistically if time allows, and by stating this scope explicitly rather than overclaiming.

- **RL training instability:** REINFORCE on LM-conditioned generation can fail to converge; mitigated by variance-reduction (baseline subtraction, entropy bonus) and a non-RL best-of-N fallback.

- **Hardware constraints:** Limited to a small quantized frozen model on consumer GPU/NPU hardware; mitigated by keeping all trainable components small and rollouts short.

- **Ethical considerations:** The method optimizes agents to evade a third-party observer, which has a benign framing here (topic-hiding as a coordination problem) but could generalize to misuse; the project will not release trained controller weights, only the method and aggregate results.

References

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